Project ID: INT10

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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

B. Tech. Project

"AI-Powered Meeting Lifecycle Management & Workflow Automation System"

Monthly Progress Report – August 2026

Under the Guidance of

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Group Members

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1. Work Distribution: This is a solo project and all the activities like requirement analysis, literature review, system design, core module development testing and deploying are all being carried out by Bontha Mallikarjun Reddy.

2. Introduction

To begin with Most of the meetings remain central to how organizations plan, collaborate and make decisions yet the way they are typically managed is far from efficient. Notes get taken inconsistently, action items go undocumented and follow really happened in any structured way. So this results in most of the useful information getting not noted. Uh in market there is ma uh many number of AI meeting assistants already accessed to help with summarisation and trans transcription, but mostly stop there like they do much to connect the information they extract to the actual workflows of the teams like assigning tasks, tracking decisions, scheduling follow ups and so on this project sets out to build a platform that goes further then a raw meeting discussions and turns them into a structured, actionable workflows by combining meeting intelligence with task management, ongoing context memory, automation.

3. Problem Statement

The problem statement is to build an AI powered meeting life cycle management and workflow automation Platform that can automatically a capture meeting discussions, generate summaries, track decisions across meetings, flag deadlines And surface contextual insights these improve collaboration accountability and productivity.

4. Objectives

This project aims to build a system that can:

- Convert meeting audio into accurate text transcriptions automatically.
- Generate summaries that are concise and genuinely reflect the context of the discussion.
- Identify the key decisions made during a meeting.
- Extract actionable items and figure out who's responsible for each one.
- Detect deadlines and automatically create reminders and calendar events for them.
- Retain context across multiple meetings so users have a searchable, connected record rather than isolated notes.
- Generate agendas for upcoming meetings based on what's still unresolved from previous ones.
- Present system analytics and task status through an interactive dashboard.

5. Literature Review

The following works and existing tools were reviewed to establish the methodological foundation for this project and to identify where current solutions fall short:

Sr. No.	Author(s)	Year	Paper / Study	Methodology / Technology	Key Findings	Research Gap
1	Vaswani et al.	2017	*Attention Is All You Need*	Transformer Architecture	Introduced the Transformer model, significantly improving NLP tasks such as summarization and text generation.	Does not address meeting-specific understanding or workflow automation.
2	Radford et al. (OpenAI)	2022	*Whisper: Robust Speech Recognition via Large-Scale Weak Supervision*	Deep Learning-based Automatic Speech Recognition (ASR)	Achieved high speech-to-text accuracy across multiple languages and noisy environments.	Focuses only on transcription without extracting meeting insights or actions.
3	Jurafsky & Martin	2023	*Speech and Language Processing*	Natural Language Processing (NLP), Named Entity Recognition, Information Extraction	Provides comprehensive techniques for extracting entities, summarization, and semantic understanding from text.	General NLP concepts; does not provide an end-to-end meeting intelligence solution.
4	Microsoft	2023	*Microsoft 365 Copilot*	Large Language Models (LLMs), Microsoft Graph Integration	Generates meeting summaries, highlights key discussion points, and suggests follow-up actions within Microsoft Teams.	Primarily works within the Microsoft ecosystem and offers limited customizable workflow automation.
5	Otter.ai Research Team	2023	*AI-Powered Meeting Assistant*	Automatic Speech Recognition, NLP, Cloud-based AI	Provides real-time transcription, searchable meeting notes, and meeting summaries.	Limited long-term context management and project lifecycle tracking across meetings.
6	Fireflies.ai	2023	*Meeting Intelligence Platform*	Speech Recognition,	Automates note-taking, extracts action items, and	Focuses mainly on transcription and summaries;

				NLP, AI Automation	integrates with productivity tools.	lacks advanced contextual memory and intelligent agenda generation.
7	Zoom Video Communications	2024	*Zoom AI Companion*	Generative AI, NLP, Meeting Intelligence	Automatically summarizes meetings and assists users with post-meeting information.	Primarily designed for meeting summarization with limited workflow tracking beyond meetings.
8	Lewis et al.	2020	*Retrieval-Augmented Generation (RAG) for Knowledge-Intensive NLP Tasks*	Retrieval-Augmented Generation (RAG), Dense Passage Retrieval	Demonstrated how combining retrieval with LLMs improves factual accuracy and contextual responses.	Not specifically applied to maintaining long-term meeting memory or workflow management.
9	Brown et al.	2020	*Language Models are Few-Shot Learners (GPT-3)*	Large Language Models (LLMs), Few-shot Learning	Showed that large language models can perform diverse language understanding and generation tasks with minimal task-specific training.	Does not address structured extraction of meeting decisions, deadlines, or enterprise workflows.

6. Research and Technology Gap

So there are many AI meeting assistance which already are available in the market which rewards around basic transcription and summarization What they generally lack is a real memory context which gives it the ability to track the project across multiple meetings and a way to turn extracted information into an active workflow automation So this is a huge gap in the current products. This project tries to close the gap By combining open source models with a workflow automation which aims to create a platform that stays context aware continuously rather than treating a meeting is an isolated event.

7. Proposed Solution (System Workflow)

The platform will turn unstructured meeting discussions into actionable workflows through the following steps:

- **Audio Capture & Transcription:** Record or accept uploaded meeting audio and convert it to text using the Whisper speech-to-text model.
- **Intelligent Summarization:** Use a large language model to analyze the transcript and produce a concise, relevant summary.
- **Insight Extraction:** Automatically identify action items, responsible participants, key decisions, and deadlines.
- **Contextual Memory:** Store meeting data so it can be searched later and so context carries over across a series of related meetings.
- **Workflow Automation:** Generate reminders and calendar events automatically based on the deadlines detected.
- **Agenda Generation:** Build agendas for upcoming meetings by looking at what's still unresolved or pending.
- **Centralized Dashboard:** Give users one place to track task status, meeting summaries, and overall progress.

8. Requirement Analysis

Functional Requirements:

- User authentication and management
- Meeting audio upload and speech-to-text conversion
- AI-generated summaries and extraction of action items, decisions, and deadlines
- Semantic search across meetings
- Reminder generation and calendar integration
- Analytics dashboard

Non-Functional Requirements:

- High accuracy and reliability in transcription and extraction

- Scalability and strong security
- Good performance and a user-friendly interface

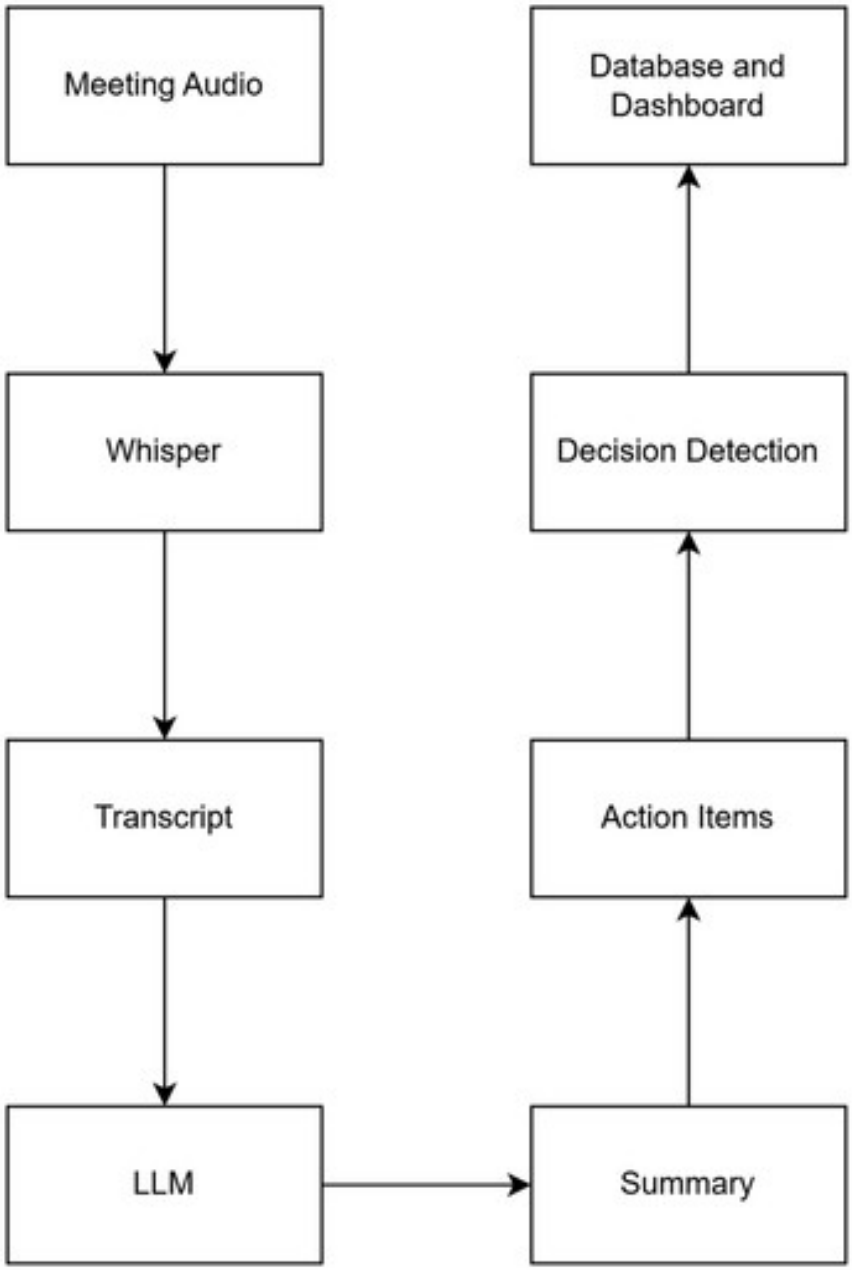
9. Technology Stack

To keep the solution customizable and privacy-conscious, the following technologies were chosen, most of them open-source:

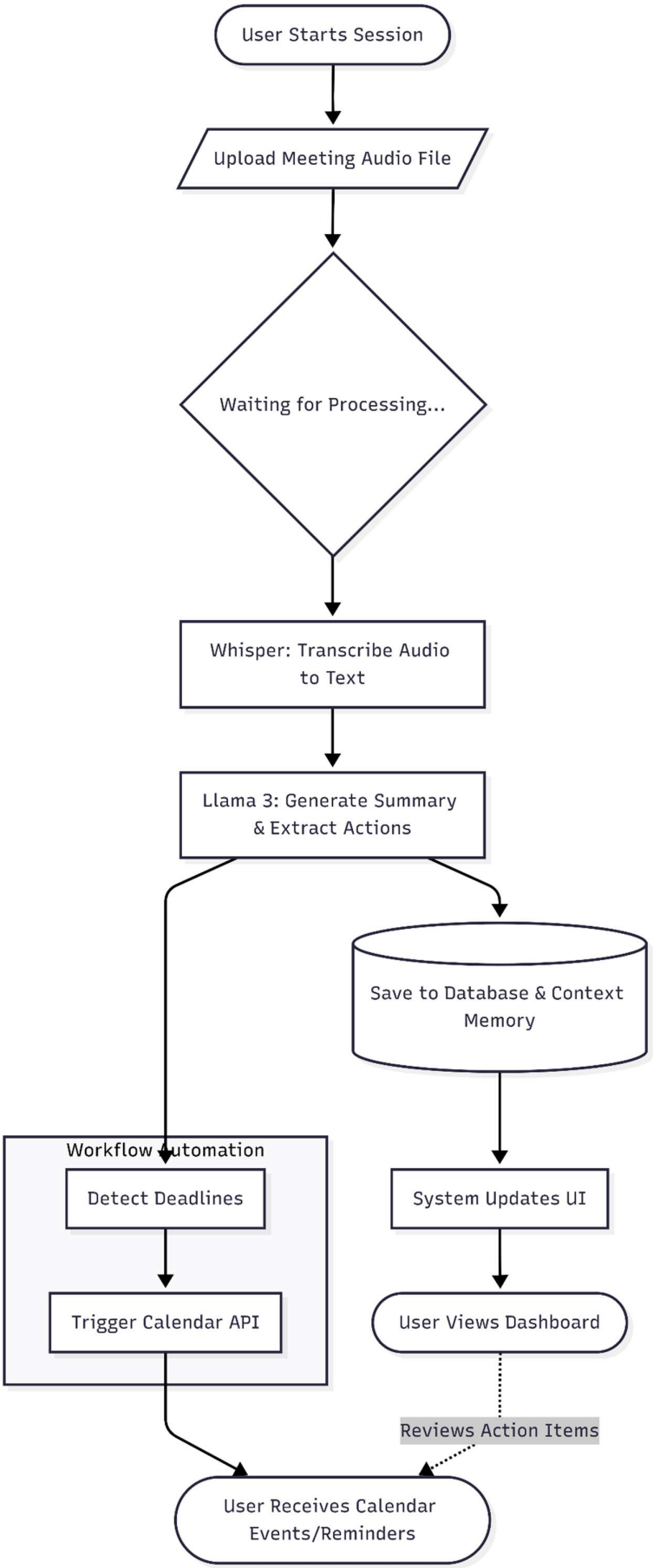
Category	**Technology**
Frontend	React.js, Tailwind CSS
Backend	FastAPI (Python)
AI/ML	Llama 3 / Mistral / OpenAI GPT (optional)
Database	PostgreSQL
Context Memory	ChromaDB
Authentication	JWT
Integrations	Google Calendar API, Microsoft Graph API, SMTP/SendGrid
Version Control	Git, GitHub
Deployment	Docker, Azure/AWS/Render
Testing	Postman
IDE	Visual Studio Code

10. System Design

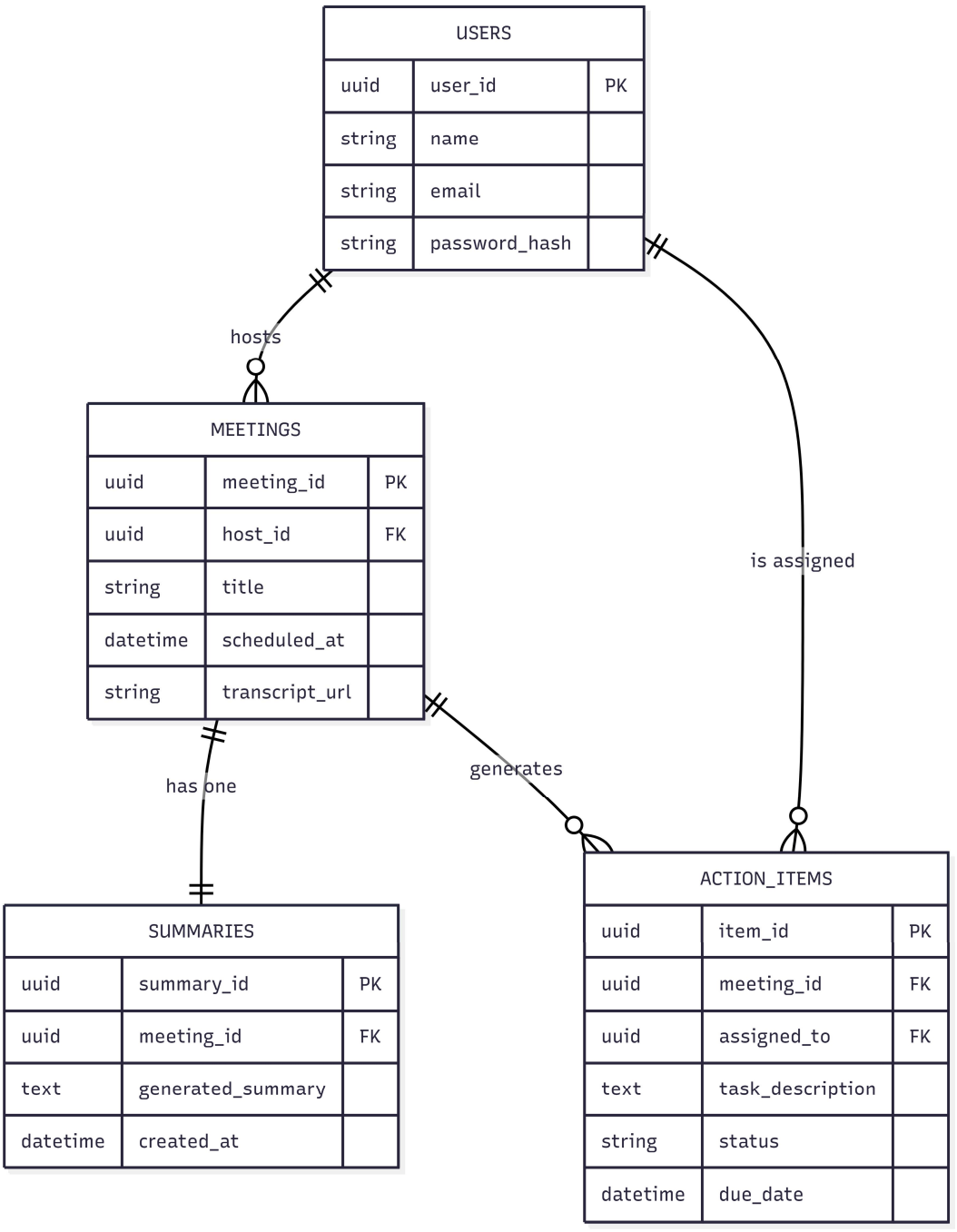
- **10.1 System Architecture / Block Diagram:**



- **10.2 Workflow / Activity Diagram:**



- **10.3 Database Schema / ER Diagram:**



11. Development and Implementation Status

At this point in the CA-2 reporting period, the project is still in the architectural planning and design phase.

- **Current Progress:** The system architecture has been finalized, module planning is complete, and the core technology stack — including the decision to use Llama 3 and FastAPI — has been settled. Preliminary UI wireframes and frontend structural planning are underway.
- **Next Steps:** The next phase involves setting up the backend repository, initializing the PostgreSQL and ChromaDB databases, and writing the API routes ahead of integrating the AI models. Actual code implementation for the core AI modules hasn't started yet.

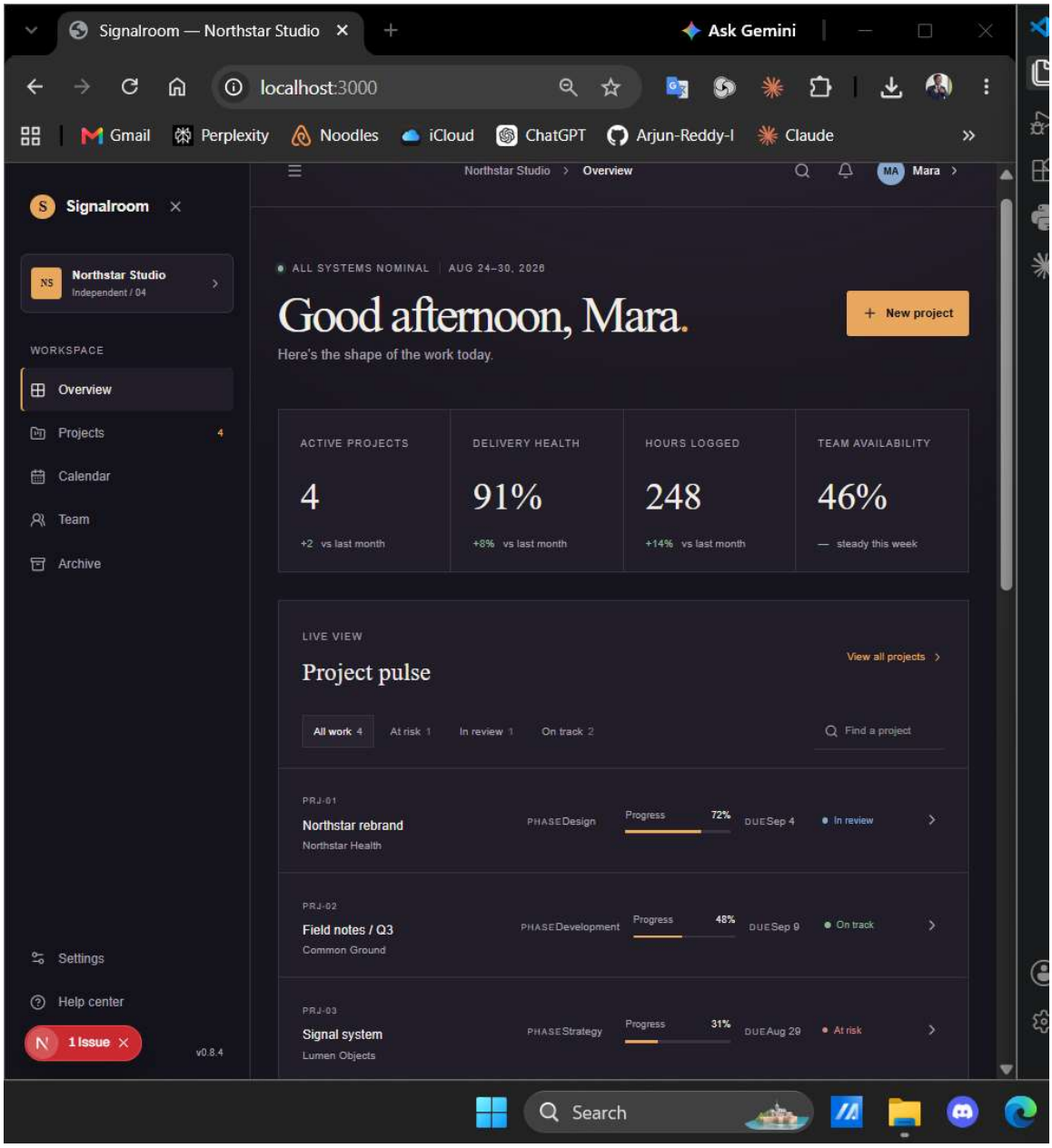
12. Testing and Debugging

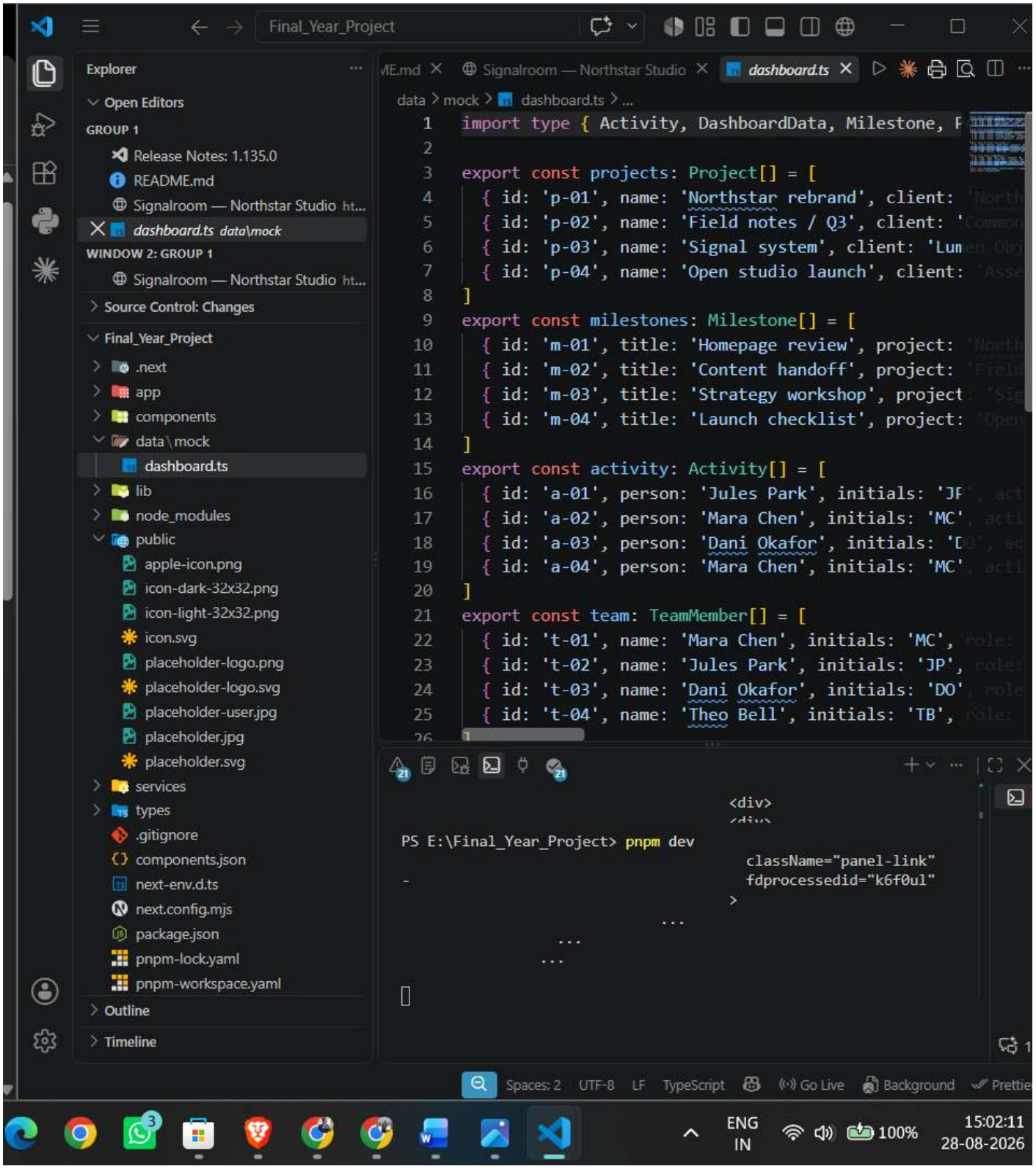
Implementation Status: Planned

A full testing suite is planned for after development is further along. Evaluation will rely on Precision, Recall, and F1 Score for the AI extraction tasks, along with response time metrics. Planned testing phases include:

- **Unit Testing:** Checking transcription accuracy and testing individual endpoints in isolation.
- **Integration Testing:** Validating action item extraction, reminder generation, and calendar event creation as they work together.
- **Functional Testing:** Testing whether context is retrieved correctly across multiple stored meetings.

UI Screenshots {On Progress}:




Final_Year_Project
Explorer
Open Editors
GROUP 1
Release Notes: 1.135.0
README.md
Signalroom — Northstar Studio
dashboard.ts data\mock
WINDOW 2: GROUP 1
Signalroom — Northstar Studio
Source Control: Changes
Final_Year_Project
.next
app
components
data \ mock
dashboard.ts
lib
node_modules
public
apple-icon.png
icon-dark-32x32.png
icon-light-32x32.png
icon.svg
placeholder-logo.png
placeholder-logo.svg
placeholder-user.jpg
placeholder.jpg
placeholder.svg
services
types
.gitignore
components.json
next-env.d.ts
next.config.mjs
package.json
pnpm-lock.yaml
pnpm-workspace.yaml
Outline
Timeline
Signalroom — Northstar Studio
dashboard.ts
data > mock > dashboard.ts > ...
import type { Activity, DashboardData, Milestone, P
export const projects: Project[] = [
{ id: 'p-01', name: 'Northstar rebrand', client:
{ id: 'p-02', name: 'Field notes / Q3', client:
{ id: 'p-03', name: 'Signal system', client: 'Lun
{ id: 'p-04', name: 'Open studio launch', client:
]
export const milestones: Milestone[] = [
{ id: 'm-01', title: 'Homepage review', project:
{ id: 'm-02', title: 'Content handoff', project:
{ id: 'm-03', title: 'Strategy workshop', project
{ id: 'm-04', title: 'Launch checklist', project:
]
export const activity: Activity[] = [
{ id: 'a-01', person: 'Jules Park', initials: 'JP
{ id: 'a-02', person: 'Mara Chen', initials: 'MC'
{ id: 'a-03', person: 'Dani Okafor', initials: 'D
{ id: 'a-04', person: 'Mara Chen', initials: 'MC'
]
export const team: TeamMember[] = [
{ id: 't-01', name: 'Mara Chen', initials: 'MC',
{ id: 't-02', name: 'Jules Park', initials: 'JP',
{ id: 't-03', name: 'Dani Okafor', initials: 'DO'
{ id: 't-04', name: 'Theo Bell', initials: 'TB',

PS E:\Final_Year_Project> pnpm dev
className="panel-link"
fdprocessedid="k6f0ul"
>
Spaces: 2
UTF-8
LF
TypeScript
Go Live
Background
ENG
IN
100%
15:02:11
28-08-2026

13. Project Plan and Progress

The timeline below reflects a slight delay in finalizing the system design and UI, which is currently being actively worked on.

Phase	Planned Duration	Current Status	Remarks
Requirement Analysis & Literature Review	Week 1	✅ Complete	Scope, requirements, and literature finalized.
System Design & UI Development	Week 2–4	🟡 In Progress	Architecture finalized; UI wireframes and schemas actively being developed.
Core Feature Development	Week 5–7	⏳ Planned	Pending backend setup and AI integration.
Dashboard Dev & Refinement	Week 8	⏳ Planned	–
Testing & Validation	Week 9–10	⏳ Planned	–
Documentation & Deployment	Week 11–12	⏳ Planned	–

14. Project Outcome

Expected Outcomes: Once complete, the system should automate meeting documentation, cut down on manual effort, and make task accountability easier to enforce. It's expected to track decisions across meetings, generate reminders and agendas automatically, and provide analytics that support better productivity.

Current Outcome/Progress So Far: So far, the project has a clearly defined problem space, a completed literature review, a finalized system architecture, and a settled technology stack built on open-source tools. Interface design and structural planning are actively underway as preparation for the core development phase.

15. Conclusion

This project shows how AI can go beyond basic meeting summarization to support the full lifecycle of meeting management. By bringing together transcription, Llama 3-powered summarization, context memory, and workflow automation, the platform aims to cut down administrative overhead and make collaboration across an organization more effective. The design work is now complete, and active development is set to begin in the next phase.

16. References

- Vaswani, A., et al., "Attention Is All You Need," NeurIPS, 2017.
- Radford, A., et al., "Whisper: Robust Speech Recognition via Large-Scale Weak Supervision," OpenAI, 2022.
- Jurafsky, D., & Martin, J. H., Speech and Language Processing, 3rd ed., 2023.
- Brown, T., et al., "Language Models are Few-Shot Learners," NeurIPS, 2020.
- Lewis, P., et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP," NeurIPS, 2020.
- Microsoft. "Microsoft 365 Copilot Documentation."
- Fireflies.ai Product Documentation.
- Otter.ai Product Documentation.
- Zoom AI Companion Documentation.

17. Similarity Report

(To be provided by Guide)

18. AI Plagiarism Report

(To be provided by Guide)